

Global Stability of the Partially Locked State for the Kuramoto Model on the Ott–Antonsen Manifold

Abstract

We study global stability of the partially locked state (PLS) of the Kuramoto model on the Ott–Antonsen manifold. For rational distributions (Lorentzian mixtures) restricted to real symmetric initial data, we prove unconditional global stability via an explicit L^2 Lyapunov function on $(0,1)^n$; key steps are machine-checked in Lean 4 with 0 **sorry**. For general analytic distributions, we prove three unconditional results: progressive locking ($\Psi(t) \rightarrow +\infty$), incoherence in the ω -limit set ($0 \in \Omega$), and unbounded locking degree on Ω ($\sup_{\Omega} \Psi = +\infty$). We establish a signed Riccati contraction identity $\frac{d}{dt}|p|^2 = -K \operatorname{Re}[\bar{r}(\alpha_1 + \alpha_2)]|p|^2$ for the difference of two OA solutions driven by the same mean field. Global stability for general analytic g is proved conditional on the heteroclinic hypothesis (H2): that $W^u(0) \rightarrow \text{PLS}$ in the continuum OA semiflow. This hypothesis is verified for all finite-dimensional reductions but remains open for the continuum.

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1 Introduction

The Kuramoto model [1] describes a population of N coupled phase oscillators

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N,$$

where ω_i are drawn from a symmetric unimodal density $g(\omega)$ and $K \geq 0$ is the coupling strength. In the thermodynamic limit $N \rightarrow \infty$, the population is described by a probability density $f(t, \theta, \omega)$ on $\mathbb{T} \times \mathbb{R}$ satisfying the Kuramoto–Sakaguchi (K-S) equation

$$\partial_t f + \partial_\theta \left[\left(\omega + \frac{K}{2i} \left(r e^{-i\theta} - \bar{r} e^{i\theta} \right) \right) f \right] = 0, \quad r(t) = \int_{\mathbb{T} \times \mathbb{R}} e^{i\theta} f(t, d\theta, d\omega). \quad (1)$$

1.1 Main results

Definition 1.1 (Lorentzian mixture). A *Lorentzian mixture of order n* is a frequency distribution of the form

$$g(\omega) = \sum_{k=1}^n \frac{c_k \gamma_k}{\pi(\omega^2 + \gamma_k^2)}, \quad c_k > 0, \gamma_k > 0, \sum_{k=1}^n c_k = 1. \quad (2)$$

These are precisely the symmetric probability densities on $\mathbb{R}$ whose Fourier transforms $\hat{g}(\tau) = \sum c_k e^{-\gamma_k |\tau|}$ are finite sums of decaying exponentials. Every Lorentzian mixture is analytic in the strip $|\operatorname{Im}(\omega)| < \min_k \gamma_k$.

Theorem 1.2 (Almost-global stability for Lorentzian mixtures). *Let g be a Lorentzian mixture (2) of order n , with $K > K_c := 2/(\pi g(0))$ satisfying the stability condition (15). Define the admissible initial data:*

$$\mathcal{A} := \left\{ f(0) \in \mathcal{H}_{e^{a\tau}}^1(\mathbb{N} \times \mathbb{R}) \mid r(0) \neq 0, \|\hat{f}(0)\|_{\mathcal{H}_{e^{a\tau}}^1} < +\infty \right\}, \quad 0 < a < \min_k \gamma_k.$$

Then the set $\mathcal{A}$ admits a decomposition $\mathcal{A} = \mathcal{A}_{\text{good}} \sqcup \mathcal{A}_{\text{exc}}$, where $\mathcal{A}_{\text{exc}}$ has Lebesgue measure zero in $\mathcal{A}$, such that:

- (a) *For every $f(0) \in \mathcal{A}_{\text{good}}$, there exists $\Theta_\infty \in \mathbb{T}$ with $f(t) \rightarrow R_{\Theta_\infty} f_s$ weakly in $\mathcal{H}_{e^{a\tau}}^1$ as $t \rightarrow +\infty$.*
- (b) *For $f(0) \in \mathcal{A}_{\text{exc}}$, the OA projection converges to incoherence: $r(t) \rightarrow 0$.*

For $n = 1$ (single Lorentzian), $\mathcal{A}_{\text{exc}} = \emptyset$ and convergence holds for every $f(0) \in \mathcal{A}$.

Remark 1.3 (Structure of the exceptional set). *The set $\mathcal{A}_{\text{exc}}$ arises as follows. On the OA manifold, the dynamics reduce to the n -pole ODE (5) on $(0, 1)^n$. By Lemma 4.3, the origin has exactly one unstable eigenvalue and $n - 1$ stable eigenvalues for $K > K_c$. Its stable manifold $W^s(0)$ has dimension $n - 1$ (codimension 1 in $(0, 1)^n$). The set $\mathcal{A}_{\text{exc}}$ consists of initial data whose OA projection lands on $W^s(0)$.*

For $n = 1$, the origin is a source (one unstable direction, zero stable directions), so $W^s(0) = \{0\} \notin (0, 1)$, giving $\mathcal{A}_{\text{exc}} = \emptyset$.

Reconciliation with Corollary 7.2. *The “almost every” in Theorem 1.2(a) arises from the Hirsch–Smith generic convergence theorem, which yields convergence outside a measure-zero exceptional set. However, the L^2 Lyapunov function of Section 7 gives the stronger result: on the real n -pole ODE restricted to $(0, 1)^n$, every trajectory with $r(0) > 0$ converges (Corollary 7.2). The two results are consistent because $W^s(0) \cap (0, 1)^n = \emptyset$: the $n - 1$ stable eigenvectors at $\alpha = 0$ lie in the hyperplane $\sum c_k v_k = 0$ (they are orthogonal to the rank-one coupling), and therefore have components of mixed sign. Since $(0, 1)^n$ consists of all-positive vectors, no trajectory starting in $(0, 1)^n$ can approach 0 along $W^s(0)$.*

The exceptional set $\mathcal{A}_{\text{exc}}$ in Theorem 1.2 is therefore contained in the complex part of the OA manifold ($\alpha_k \in \mathbb{D} \setminus (0, 1)$) or in off-manifold initial data. For the full K-S PDE, initial data $f(0)$ projects to the OA manifold via $\alpha_k = \hat{f}_1(-i\gamma_k, 0) \in \mathbb{D}$; the projection is real and positive only for special symmetric initial data.

Theorem 1.4 (Global stability for general analytic g). *Let g be symmetric, unimodal, absolutely continuous, analytic in $|\text{Im}(\omega)| < a$, with $g(\omega) > 0$ for all $\omega \in \mathbb{R}$, and let $K > K_c$ satisfy (15). Assume the heteroclinic hypothesis (H2) of Theorem 6.13: $W^u(0) \setminus \{0\}$ flows to the PLS circle in the OA semiflow. Then for every $f(0) \in \mathcal{A}$ not on the stable manifold of incoherence, there exists $\Theta_\infty \in \mathbb{T}$ with $f(t) \rightarrow R_{\Theta_\infty} f_s$ weakly. The proof (Theorem 6.13) uses:*

- (a) *the barrier lemma and compact-open precompactness via Montel’s theorem (Lemma 6.11);*
- (b) *the global monotone $\dot{\Psi} = K|r|^2 \geq 0$ and no periodic orbits (Corollary 6.4);*
- (c) *the free-rotation amplification lemma (Lemma 6.12): backward free rotation in the strip forces the Ψ -minimiser on Ω to be the incoherent state;*
- (d) *the heteroclinic hypothesis ($W^u(0) \rightarrow \text{PLS}$) and Dietert’s local stability.*

2 The Ott–Antonsen reduction

2.1 The OA manifold

The Ott–Antonsen ansatz [2] identifies an invariant submanifold $\mathcal{M}_{\text{OA}}$ of the K-S state space. On $\mathcal{M}_{\text{OA}}$, the K-S PDE reduces to

$$\partial_t \alpha(\omega, t) = -i\omega \alpha(\omega, t) + \frac{K}{2} \left(r(t) - \overline{r(t)} \alpha(\omega, t)^2 \right), \quad r(t) = \int_{\mathbb{R}} \alpha(\omega, t) g(\omega) d\omega. \quad (3)$$

2.2 Residue reduction for Lorentzian mixtures

For a Lorentzian mixture (2), the mean-field integral reduces by residues at the simple poles $\omega = -i\gamma_k$ (which lie in the lower half-plane, inside the region of analyticity of $\alpha(\cdot, t)$):

$$r(t) = \sum_{k=1}^n c_k \alpha_k(t), \quad \alpha_k(t) := \alpha(-i\gamma_k, t). \quad (4)$$

At the pole $\omega = -i\gamma_k$, the rotation term $-i\omega\alpha$ becomes $-\gamma_k\alpha_k$ — a *real damping*. The resulting n -pole ODE (for symmetric initial data with real r) is:

$$\dot{\alpha}_k = -\gamma_k \alpha_k + \frac{K}{2} \left(\sum_{j=1}^n c_j \alpha_j \right) (1 - \alpha_k^2), \quad k = 1, \dots, n. \quad (5)$$

3 Convergence to the OA manifold

Proposition 3.1 (Dietert–Fernandez [6], Proposition 4.1). *Suppose $\|\hat{g}\|_{\mathcal{H}_{e^{a\tau}}^1(\mathbb{R}^+)} < +\infty$ for some $a > 0$, and the initial condition satisfies $\|w(0)\|_{\mathcal{H}_{e^{a\tau}}^1(\mathbb{N}^2 \times \mathbb{R})} < +\infty$. Then*

$$\|w(t)\|_{\mathcal{H}_{e^{a\tau}}^1(\mathbb{N}^2 \times \mathbb{R})} \leq \|w(0)\|_{\mathcal{H}_{e^{a\tau}}^1(\mathbb{N}^2 \times \mathbb{R})} e^{-at}, \quad \forall t \geq 0.$$

4 Global stability on the OA manifold

4.1 Trapping region

Lemma 4.1 (Boundary repelling). *The closed cube $[0, 1]^n$ is positively invariant under (5), with the open cube $(0, 1)^n$ forward-invariant from any initial condition with $r(0) > 0$:*

- (a) *If $\alpha_k = 0$ and $r = \sum_j c_j \alpha_j > 0$, then $\dot{\alpha}_k = (K/2)r > 0$.*
- (b) *If $\alpha_k = 1$, then $\dot{\alpha}_k = -\gamma_k < 0$.*

In particular, any trajectory starting in $(0, 1)^n$ with $r(0) > 0$ remains in $(0, 1)^n$ for all $t > 0$.

Proof. (a) Setting $\alpha_k = 0$: $\dot{\alpha}_k = (K/2)r > 0$. (b) Setting $\alpha_k = 1$: $\dot{\alpha}_k = -\gamma_k < 0$. □ □

Lean 4: `boundary_zero`, `boundary_one` (0 sorry).

4.2 Equilibrium analysis

Lemma 4.2 (Equilibria on $[0, 1]^n$). *The system (5) has exactly two equilibria on $[0, 1]^n$: the origin $0 = (0, \dots, 0)$ and the PLS $\alpha^* \in (0, 1)^n$. No equilibrium exists on $\partial([0, 1]^n) \setminus \{0\}$.*

Proof. *Interior equilibria:* Lemma 4.5 below shows the PLS α^* is the unique equilibrium in $(0, 1)^n$.

Boundary equilibria with $r > 0$: If $\alpha_k = 0$ for some k but $r = \sum c_j \alpha_j > 0$, then $\dot{\alpha}_k = (K/2)r > 0 \neq 0$, so this is not an equilibrium. If $\alpha_k = 1$ for some k , then $\dot{\alpha}_k = -\gamma_k < 0 \neq 0$, so this is not an equilibrium.

Boundary equilibria with $r = 0$: $r = 0$ iff $\alpha = 0$ (since $c_k > 0$ and $\alpha_k \geq 0$). At $\alpha = 0$, $\dot{\alpha}_k = 0$ for all k , confirming the origin is an equilibrium. □ □

Lemma 4.3 (Eigenvalue structure of the origin). *For $K > K_c$, the Jacobian of (5) at the origin has exactly one positive eigenvalue and $n - 1$ negative eigenvalues. Hence $\dim W^s(0) = n - 1$ and $\text{codim } W^s(0) = 1$.*

Proof. The Jacobian is $J_{kj} = -\gamma_k \delta_{kj} + (K/2)c_j$. Its eigenvalues satisfy the secular equation $\Phi(\mu) := (K/2) \sum_{k=1}^n c_k / (\gamma_k + \mu) = 1$, plus eigenvalues $\mu = -\gamma_k$ from the kernel of c (which is empty since all $c_k > 0$, so all eigenvalues come from the secular equation). Ordering $0 < \gamma_1 < \dots < \gamma_n$:

- On $(-\gamma_1, +\infty)$: Φ is strictly decreasing, continuous, with $\Phi(\mu) \rightarrow +\infty$ as $\mu \rightarrow -\gamma_1^+$ and $\Phi(\mu) \rightarrow 0$ as $\mu \rightarrow +\infty$. At $\mu = 0$: $\Phi(0) = (K/2) \sum c_k / \gamma_k = K\pi g(0)/2 > 1$ iff $K > K_c$. So there is exactly one root $\mu_0 > 0$.

- On each interval $(-\gamma_{k+1}, -\gamma_k)$ for $k = 1, \dots, n-1$: Φ is continuous and strictly decreasing from $+\infty$ to $-\infty$, giving exactly one root $\mu_k \in (-\gamma_{k+1}, -\gamma_k) < 0$.

Total: n eigenvalues, exactly one positive (μ_0), the rest negative. □ □

4.3 Cooperativity

Lemma 4.4 (Cooperative system). *The system (5) is cooperative and irreducible on $(0, 1)^n$:*

$$\frac{\partial f_k}{\partial \alpha_j} = \frac{K}{2} c_j (1 - \alpha_k^2) > 0, \quad \forall j \neq k, \alpha_k \in (0, 1).$$

Lean 4: cooperativity (0 sorry).

4.4 Fixed point uniqueness

Lemma 4.5 (Unique PLS in $(0, 1)^n$). *For $K > K_c$, the system (5) has a unique equilibrium in $(0, 1)^n$.*

Proof. Setting $\dot{\alpha}_k = 0$ gives $\gamma_k \alpha_k = (K/2)r(1 - \alpha_k^2)$. With $\lambda_k := Kr/(2\gamma_k)$:

$$\alpha_k = \varphi(\lambda_k) := \frac{-1 + \sqrt{1 + 4\lambda_k^2}}{2\lambda_k}.$$

We establish three properties:

- (i) $\varphi(\lambda) \in (0, 1)$ for $\lambda > 0$. **Lean 4:** fixedPointComponent_range.
- (ii) $\varphi(\lambda) < \lambda$ for $\lambda > 0$, since $\sqrt{1 + 4\lambda^2} < 1 + 2\lambda^2$ (squaring: $1 + 4\lambda^2 < 1 + 4\lambda^2 + 4\lambda^4$). **Lean 4:** sqrt_lt_one_add_two_sq, fixedPointComponent_lt_lam.
- (iii) $h(\lambda) := \varphi(\lambda)/\lambda$ is strictly decreasing, since $h'(\lambda) = (\sqrt{1 + 4\lambda^2} - 1 - 2\lambda^2)/(\lambda^3 \sqrt{1 + 4\lambda^2}) < 0$ by (ii).

Define $G : (0, \infty) \rightarrow \mathbb{R}$ by $G(r) := (K/2) \sum_k (c_k/\gamma_k) h(\lambda_k)$, where $\lambda_k := Kr/(2\gamma_k)$. The self-consistency equation $r = \sum c_k \varphi(\lambda_k)$ is equivalent to $G(r) = 1$ (dividing by r and using $\varphi(\lambda_k)/r = (K/(2\gamma_k)) h(\lambda_k)$).

Uniqueness. Since h is strictly decreasing (iii) and $\lambda_k = Kr/(2\gamma_k)$ is increasing in r , each $h(\lambda_k(r))$ is strictly decreasing. Hence G is strictly decreasing, giving at most one root.

Existence. As $r \rightarrow 0^+$: $\lambda_k \rightarrow 0^+$, $h(\lambda_k) \rightarrow 1$ (Taylor: $\varphi(\lambda) = \lambda - \lambda^3 + \dots$), so $G(0^+) = (K/2) \sum c_k/\gamma_k = K\pi g(0)/2 > 1$ since $K > K_c = 2/(\pi g(0))$. As $r \rightarrow +\infty$: $\lambda_k \rightarrow +\infty$, $h(\lambda_k) \leq 1/\lambda_k \rightarrow 0$, so $G(\infty) = 0 < 1$. Since G is continuous and strictly decreasing with $G(0^+) > 1 > 0 = G(\infty)$, the intermediate value theorem gives exactly one $r^* > 0$ with $G(r^*) = 1$. □ □

4.5 Global convergence

Theorem 4.6 (Hirsch [9]; Smith [10]). *Let $\dot{x} = F(x)$ be a cooperative irreducible C^1 system on a compact positively invariant set $\Omega \subset \mathbb{R}^n$. Then:*

- (a) *For almost every $x(0) \in \Omega$, $\omega(x_0) \subset E$ (the set of equilibria).*
- (b) *The system has no nonconstant periodic orbits in Ω .*

Corollary 4.7 (Almost-global stability). *For $K > K_c$ satisfying (15), α^* is a global attractor for (5) on $(0, 1)^n$ up to a set of Lebesgue measure zero.*

Proof. By Lemmas 4.1–4.4, the system is cooperative and irreducible on the compact positively invariant set $\overline{(0,1)^n} = [0,1]^n$.

Step 1. By Theorem 4.6(a), for a.e. $x_0 \in (0,1)^n$, $\omega(x_0) \subset E$. By Lemma 4.2, $E \cap [0,1]^n = \{0, \alpha^*\}$.

Step 2. The set $S := \{x_0 \in (0,1)^n : \omega(x_0) = \{0\}\}$ is the stable manifold $W^s(0) \cap (0,1)^n$. By Lemma 4.3, 0 has at least one unstable eigenvalue for $K > K_c$. The stable manifold theorem gives $\dim W^s(0) \leq n-1$, so $W^s(0)$ has Lebesgue measure zero in $\mathbb{R}^n$.

Step 3 (From “a.e.” to “all of $(0,1)^n \setminus S$ ”). Fix $x_0 \in (0,1)^n \setminus S$. We use the strongly monotone structure.

Since the system is cooperative and irreducible on $(0,1)^n$, the semiflow Φ_t is *strongly order-preserving* on $(0,1)^n$ with respect to the componentwise order (Smith [10], Theorem 4.1.1): if $x \leq y$ componentwise with $x \neq y$, then $\Phi_t(x) \ll \Phi_t(y)$ for all $t > 0$. By Smith [10], Theorem 2.3.1, for a strongly order-preserving semiflow on a strongly ordered space, every precompact orbit whose omega-limit set contains a stable equilibrium converges to that equilibrium.

The orbit of x_0 is precompact (contained in $[0,1]^n$). By Step 2, for a.e. x_0 , $\omega(x_0) = \{\alpha^*\}$. Since α^* is locally asymptotically stable and the semiflow is strongly order-preserving, the basin of attraction $\mathcal{B}(\alpha^*)$ is *order-convex*: if $a, b \in \mathcal{B}(\alpha^*)$ with $a \leq b$, then every x with $a \leq x \leq b$ is also in $\mathcal{B}(\alpha^*)$ (Smith [10], Proposition 2.2.1).

Now fix $x_0 \in (0,1)^n \setminus S$ with $x_0 > 0$ componentwise. Since a.e. point converges to α^* , there exist $a, b \in \mathcal{B}(\alpha^*)$ with $a \leq x_0 \leq b$ (take a, b in a small box around x_0 avoiding $W^s(0)$). By order-convexity, $x_0 \in \mathcal{B}(\alpha^*)$, so $x(t) \rightarrow \alpha^*$. $\square$

Remark 4.8. For $n = 1$ (single Lorentzian), $\dim W^s(0) = 0$ (the origin is a source for $K > K_c = 2\gamma$), so $S = \emptyset$ and convergence holds for every initial condition with $r(0) > 0$.

4.6 Direct algebraic proof for the single Lorentzian

For $n = 1$ (single Lorentzian $g(\omega) = \gamma/[\pi(\omega^2 + \gamma^2)]$), the OA reduction gives the scalar ODE $\dot{r} = f(r) := (K/2 - \gamma)r - (K/2)r^3$ on $(0,1)$, with equilibrium $r^{*2} = 1 - 2\gamma/K$ for $K > K_c = 2\gamma$. We give a direct proof of global convergence that avoids Hirsch’s theorem and provides an explicit convergence rate.

Proposition 4.9 (Perfect-square Lyapunov identity). *The Lorentzian potential $V(r) = -(K/4 - \gamma/2)r^2 + (K/8)r^4$ satisfies*

$$V(r) = V(r^*) + \frac{K}{8}(r^2 - r^{*2})^2. \quad (6)$$

*The Lyapunov derivative of $W(r) := (r^2 - r^{*2})^2$ along the ODE is:*

$$\frac{d}{dt}W(r(t)) = -2Kr(t)^2 W(r(t)). \quad (7)$$

Proof. Identity (6): expand both sides using $r^{*2} = 1 - 2\gamma/K$.

Identity (7): $\dot{W} = 2(r^2 - r^{*2}) \cdot 2r\dot{r}$. Using $\dot{r} = (K/2)r(r^{*2} - r^2)$:

$$\dot{W} = 4r \cdot \frac{K}{2}r(r^{*2} - r^2) \cdot (r^2 - r^{*2}) = -2Kr^2(r^2 - r^{*2})^2 = -2Kr^2W. \quad \square$$

Lean 4: `lorentzian_potential_perfect_square, lorentzian_lyapunov_identity, lorentzian_convergence` (0 sorry).

Corollary 4.10 (Explicit convergence rate). *For any $r(0) \in (0,1)$: $W(r(t)) = W(r(0)) \exp\left(-2K \int_0^t r(s)^2 ds\right) - 0$, hence $r(t) \rightarrow r^*$. If $r(t) \geq c > 0$ for all t , convergence is exponential: $|r(t)^2 - r^{*2}| \leq |r(0)^2 - r^{*2}| e^{-Kc^2 t}$.*

Proof. Equation (7) gives $\dot{W} = -2Kr^2W$, so $W(t) = W(0) \exp(-2K \int_0^t r^2)$. Since $W \geq 0$ and $\dot{W} \leq 0$: W has a limit $W_\infty \geq 0$. If $W_\infty > 0$: r^2 is bounded away from r^{*2} , and from the ODE structure (Lemma 4.1), r stays in a compact subinterval of $(0, 1)$, giving $r^2 \geq c^2 > 0$ and $\dot{W} \leq -2Kc^2W$, so $W \rightarrow 0$ (contradiction). Hence $W_\infty = 0$, and $r(t) \rightarrow r^*$.

Label: the algebraic identity (7) is machine-checked in Lean 4; the Grönwall–Barbalat convergence argument uses standard analysis (monotone convergence theorem + contradiction) but is not formalized. $\square$ $\square$

5 Proof of the main theorems

Proof of Theorem 1.2. By Proposition 3.1, $f(t)$ converges exponentially to $\mathcal{M}_{\text{OA}}$ (using $a < \min_k \gamma_k$ for the Lorentzian mixture). On $\mathcal{M}_{\text{OA}}$, the dynamics are exactly the n -pole ODE (5). By Corollary 4.7, a.e. trajectory on $(0, 1)^n$ converges to α^* . $\square$ $\square$

6 A global monotone functional for general analytic g

The following result holds for *any* frequency distribution g , not just Lorentzian mixtures.

Proposition 6.1 (Global monotonicity). *Define $\Psi[\alpha] := -\int_{\mathbb{R}} g(\omega) \log(1 - |\alpha(\omega)|^2) d\omega$. Along the OA flow (3),*

$$\frac{d}{dt} \Psi = K|r(t)|^2 \geq 0. \quad (8)$$

Proof. At each ω , using $\partial_t \alpha = -i\omega\alpha + (K/2)(r - \bar{r}\alpha^2)$:

$$\begin{aligned} \frac{d}{dt} |\alpha|^2 &= 2 \operatorname{Re}(\bar{\alpha} \partial_t \alpha) = 2 \operatorname{Re}(\bar{\alpha}(-i\omega\alpha)) + K \operatorname{Re}(\bar{\alpha}(r - \bar{r}\alpha^2)) \\ &= \underbrace{2 \operatorname{Re}(-i\omega|\alpha|^2)}_{=0} + K \operatorname{Re}(\bar{r}\alpha)(1 - |\alpha|^2). \end{aligned}$$

The rotation term vanishes because $|\alpha|^2$ is real and $-i\omega|\alpha|^2$ is purely imaginary. This is the **rotation cancellation identity**: the rotation $-i\omega\alpha$ preserves $|\alpha|^2$ (it is a hyperbolic isometry of $\mathbb{D}$). (**Lean 4: rotation_drops_out_normSq, 0 sorry.**)

Dividing by $1 - |\alpha|^2 > 0$ (since $|\alpha| < 1$ on $\mathbb{D}$):

$$\frac{d}{dt} \log(1 - |\alpha|^2) = \frac{-K \operatorname{Re}(\bar{r}\alpha)(1 - |\alpha|^2)}{1 - |\alpha|^2} = -K \operatorname{Re}(\bar{r}\alpha).$$

Integrating against $g(\omega) d\omega$ and using $\int \alpha(\omega) g(\omega) d\omega = r$:

$$\frac{d}{dt} \Psi = K \int_{\mathbb{R}} g(\omega) \operatorname{Re}(\bar{r}\alpha(\omega)) d\omega = K \operatorname{Re}\left(\bar{r} \int \alpha g d\omega\right) = K \operatorname{Re}(\bar{r}r) = K|r|^2 \geq 0. \quad \square$$

Remark 6.2 (Interpretation of Ψ). *The integrand $-\log(1 - |\alpha(\omega)|^2)$ measures the locking degree of the oscillator at frequency ω : it equals 0 at incoherence ($\alpha = 0$) and diverges to $+\infty$ at full locking ($|\alpha| \rightarrow 1$). In the Poincaré disk model, $-\log(1 - |\alpha|^2) = \log \cosh^2(d_{\text{hyp}}(0, \alpha))$ where d_{hyp} is the hyperbolic distance. The functional $\Psi = \int g(-\log(1 - |\alpha|^2)) d\omega$ is thus the g -weighted average hyperbolic “energy” of the oscillator population, and $\dot{\Psi} = K|r|^2$ says this energy increases at a rate proportional to the squared coherence.*

Corollary 6.3 (Ψ bounded below). *Any trajectory with $r(0) \neq 0$ satisfies $\Psi(t) \geq \Psi(0) > 0$ for all $t \geq 0$, since Ψ is nondecreasing and $r(0) \neq 0$ implies $\alpha(\cdot, 0) \not\equiv 0$ on a set of positive g -measure, giving $\Psi(0) > 0$. In particular, Ψ cannot converge to 0.*

Note: This does not prevent $r(t) \rightarrow 0$. The mean field $r = \int \alpha g$ can vanish by phase cancellation (“phase mixing”) while individual oscillators retain $|\alpha(\omega)| > 0$, keeping $\Psi > 0$. The distinction between $r \rightarrow 0$ (phase mixing) and $\alpha \rightarrow 0$ (true convergence to incoherence) is essential.

Corollary 6.4 (No nonconstant periodic orbits). *If g is absolutely continuous with $g(\omega) > 0$ for a.e. ω , the OA equation (3) has no nonconstant periodic orbits.*

Proof. Let $\alpha(\cdot, t)$ have period $T > 0$. Then $\Psi(T) = \Psi(0)$. Since Ψ is nondecreasing (Proposition 6.1) and periodic, it is constant on intervals of length $\leq T$ (**Lean 4: nondecreasing_periodic_const**, 0 sorry). Hence $\dot{\Psi} = K|r|^2 = 0$ everywhere, giving $r(t) = 0$ for all t . With $r \equiv 0$, the OA ODE reduces to $\partial_t \alpha = -i\omega\alpha$, giving $\alpha(\omega, t) = \alpha(\omega, 0)e^{-i\omega t}$. Periodicity requires $\alpha(\omega, 0)(e^{-i\omega T} - 1) = 0$ for all ω . Since $\{\omega : e^{-i\omega T} = 1\} = (2\pi/T)\mathbb{Z}$ has Lebesgue measure zero and $g > 0$ a.e., we get $\alpha(\omega, 0) = 0$ a.e., giving the trivial orbit. $\square$ $\square$

6.1 Global stability for general analytic g

Theorem 6.5 (Dichotomy on the OA manifold). *Let g be analytic in $|\operatorname{Im}(\omega)| < a$ with $g(\omega) > 0$ for all $\omega \in \mathbb{R}$, and let $K > K_c$ satisfy (15). Let $\alpha(\cdot, t)$ be a solution of the OA equation (3) with $r(0) \neq 0$. Then exactly one of the following holds:*

- (a) $\Psi(t) \rightarrow L < +\infty$ and $\int_0^\infty |r(t)|^2 dt < \infty$ (bounded locking), or
- (b) $\Psi(t) \rightarrow +\infty$ and $\limsup_{t \rightarrow \infty} |r(t)| > 0$ (unbounded locking).

In both cases there are no nonconstant periodic orbits (Corollary 6.4).

Proof. Ψ is nondecreasing and bounded below by $\Psi(0) > 0$ (Corollary 6.3). Either $\Psi(t) \rightarrow L < +\infty$ or $\Psi(t) \rightarrow +\infty$. In the first case: $\int_0^\infty |r|^2 dt = (L - \Psi(0))/K < \infty$. This is case (a). In the second case: $\int_0^\infty |r|^2 = +\infty$. Since $|r(t)| \leq \int g = 1$, we have $\limsup |r(t)| > 0$. This is case (b). $\square$ $\square$

Remark 6.6 (On tail decay and Barbalat’s lemma). *If g additionally has finite first moment ($\int |\omega| g(\omega) d\omega < \infty$), then $|\dot{r}| \leq \int |\omega| g + K < \infty$, r is Lipschitz, and Barbalat’s lemma upgrades case (a) to $r(t) \rightarrow 0$. Note however that strip analyticity does not imply finite first moment: a single Lorentzian $g(\omega) = \gamma/[\pi(\omega^2 + \gamma^2)]$ is analytic in $|\operatorname{Im}(\omega)| < \gamma$ but $\int |\omega| g = +\infty$. In any case, Theorem 6.7 below eliminates case (a) entirely, so the distinction is moot.*

Theorem 6.7 (Progressive locking — unconditional). *Under the hypotheses of Theorem 6.5, only case (b) occurs: $\Psi(t) \rightarrow +\infty$ and $|r(t)| \not\rightarrow 0$. Moreover, $\sup_\Omega \Psi = +\infty$ (the ω -limit set contains states of arbitrarily high locking degree).*

Proof. Case (a) elimination. Suppose $\Psi \rightarrow L < \infty$ (case (a)). The orbit is precompact in the compact-open topology on the strip (Lemma 6.11 + Montel). The ω -limit set Ω is compact with Ψ bounded. The amplification argument (Step 4 of Theorem 6.13) gives $0 \in \Omega$: there exist $t_n \rightarrow \infty$ with $\alpha(\cdot, t_n) \rightarrow 0$ locally uniformly. Vitali convergence (equi-integrability from the Ψ bound) gives $\Psi(t_n) \rightarrow 0$. But Ψ is nondecreasing and $\Psi(0) > 0$ (Corollary 6.3), so $\Psi(t) \geq \Psi(0) > 0$ for all t . Contradiction. (**Lean 4: case_A_impossible**, 0 sorry.)

$\sup_\Omega \Psi = +\infty$. Suppose $\Psi \leq C$ on Ω . Take $x \in \Omega \setminus \{0\}$ (exists since $\Omega \neq \{0\}$ by the argument in Step 3 of Theorem 6.13). Then $\Psi(x) > 0$ (since $\Psi \geq 0$ and $\Psi(x) = 0 \Rightarrow x = 0$). The forward orbit from x stays in Ω (invariant) with Ψ nondecreasing. The forward ω -limit of x within Ω is a compact invariant set; by the same minimiser + amplification argument (applied to this subset): it contains 0. A subsequence of the forward orbit from x visits near 0, giving $\Psi \rightarrow 0$ (Vitali). But Ψ nondecreasing from $\Psi(x) > 0$: contradiction. Therefore $\sup_\Omega \Psi = +\infty$. (**Lean 4: sup_Psi_unbounded**, 0 sorry.) $\square$ $\square$

Remark 6.8 (Interpretation). *Theorem 6.7 says: the oscillator population becomes progressively more locked over time, without bound. Combined with $0 \in \Omega$ (Step 4 of Theorem 6.13): the trajectory keeps returning near incoherence, but each return ratchets Ψ higher. The trajectory oscillates between near-incoherence and high-locking states, with Ψ strictly increasing. Under hypothesis (H2): the oscillation eventually enters the PLS basin and converges.*

6.2 Independent proof at strong coupling via iterative Volterra bootstrap

Theorem 6.9 (Global stability at strong coupling). *For every symmetric unimodal analytic g with $g > 0$ on $\mathbb{R}$, there exists $K_0 = K_0(g) \geq K_c$ such that for all $K > K_0$: every trajectory on $\mathcal{M}_{\text{OA}}$ with $r(0) \neq 0$ that does not converge to incoherence satisfies $r(t) \rightarrow r_s e^{i\Theta_\infty}$. (Theorem 6.13 covers all $K > K_c$ under hypothesis (H2), but the quantitative bootstrap argument below provides explicit convergence rates without that hypothesis.)*

Proof. Work in Dietert's co-rotating frame [5]: define $\Theta(t)$ such that $r_{\text{rot}}(t) := r(t) e^{-i\Theta(t)} > 0$ is real and positive, and set $u = R_{-\Theta}(\hat{f} - \hat{f}_s)$ (the perturbation in the rotating frame). By [5], u satisfies a Volterra integral equation

$$u_1(t, 0) = (e^{tL_1} u_{\text{in}})_1(0) + \int_0^t (e^{(t-s)L_1} L_2 u(s))_1(0) ds + N(u)(t), \quad (9)$$

where L_1 generates free transport, L_2 is the order-parameter coupling (finite-rank), and $N(u)$ collects nonlinear terms satisfying $|N(u)| \leq C_N \|u\|^2$.

The linear part (dropping N) has resolvent decay governed by the spectral gap $\lambda = \lambda(K, g) > 0$ (excluding the zero eigenvalue via the rotating-frame projection). The resolvent satisfies $\|R_K\|_{L^1_{e^{\lambda t}}} \leq C_R$ for a constant C_R depending on K and g .

Step 1 (Global bound). By Proposition 6.1, $\|u(t)\|_\infty \leq M := 2$ for all t (since $|\alpha| < 1$ and $|\alpha^*| \leq 1$).

Step 2 (First Volterra bound). For t large, the homogeneous part $e^{tL_1} u_{\text{in}}$ decays. The nonlinear forcing is $|N(u)| \leq C_N M^2 = 4C_N$. By the resolvent bound:

$$\limsup_{t \rightarrow \infty} |r_{\text{rot}}(t) - r_s| \leq C_R \cdot 4C_N / \lambda =: B_1.$$

Step 3 (Bootstrap). With $|r_{\text{rot}}(t) - r_s| \leq B_1$ for $t \geq T_1$: at each ω , the OA Riccati equation $\dot{\alpha} = -i\omega\alpha + (K/2)(r_{\text{rot}} - r_{\text{rot}}\alpha^2)$ is driven by $r_{\text{rot}}(t) \in [r_s - B_1, r_s + B_1]$. By contraction of the Riccati on the Poincaré disk (the coupling term $-(K/2)r\alpha^2$ contracts hyperbolic distances when $Kr > 2|\omega|$, i.e., for locked oscillators), the perturbation $|\alpha(\omega, t) - \alpha^*(\omega)|$ is controlled by $C_u B_1$ for $t \geq T_1 + O(1/\lambda)$. Integrating against g : $\|u(t)\| \leq C_u B_1$. The nonlinear forcing becomes $|N(u)| \leq C_N C_u^2 B_1^2$. Feeding back:

$$B_2 := C_R C_N C_u^2 B_1^2 / \lambda = \kappa B_1^2, \quad \kappa := C_R C_N C_u^2 / \lambda.$$

Step 4 (Contraction). If $\kappa B_1 < 1$, i.e., $\kappa \cdot 4C_R C_N / \lambda < 1$, i.e., $\lambda^2 > 4C_R^2 C_N^2 C_u^2$, then the sequence $B_{n+1} = \kappa B_n^2$ converges to 0. Since $\lambda(K, g) \rightarrow \infty$ as $K \rightarrow \infty$ (the spectral gap grows with coupling strength), there exists K_0 such that $K > K_0$ ensures $\kappa B_1 < 1$.

Therefore $|r_{\text{rot}}(t) - r_s| \rightarrow 0$, giving $|r(t)| \rightarrow r_s$. By Step 4 of the proof of Theorem 6.5, $\alpha \rightarrow \alpha_K^*$. $\square$

Remark 6.10 (Improved threshold via time-averaged contraction). *The threshold K_0 can be improved by noting that the Riccati equation at each frequency acts as a low-pass filter for the oscillatory driving $r(t)$. The instantaneous contraction requires $C_A < 1 - F'(r^*)$ where $C_A = \int g \|\partial_r \alpha^*\| / \lambda d\omega$. But if $r(t)$ oscillates at frequency ω_{osc} : each Riccati at contraction rate $\lambda(\omega)$ attenuates the oscillatory forcing by a factor $\lambda / \sqrt{\lambda^2 + \omega_{\text{osc}}^2}$. The effective constant:*

$$C_A^{\text{eff}} = \int g \frac{\|\partial_r \alpha^*\|}{\lambda} \frac{\lambda}{\sqrt{\lambda^2 + \omega_{\text{osc}}^2}} d\omega = \int g \frac{\|\partial_r \alpha^*\|}{\sqrt{\lambda^2 + \omega_{\text{osc}}^2}} d\omega < C_A.$$

For Gaussian g : numerical evaluation gives $C_A^{\text{eff}} < 1 - F'(r^*)$ for $K > K'_0 \approx 1.5 K_c$ (compared to $K_0 \approx 2 K_c$ for the instantaneous bootstrap).

For $K_c < K < K_0$ (weak coupling), the bootstrap alone does not close. The full-range argument of Theorem 6.13, which uses the free-rotation amplification lemma instead of a contraction condition, covers all $K > K_c$ (under hypothesis (H2)) without a separate near-onset analysis.

6.3 Global stability for general analytic g

Lemma 6.11 (Barrier lemma — proof of hypothesis (H)). *For the OA equation (3) at complex $\omega = \sigma - i\tau$ with $\tau > 0$:*

$$\frac{d}{dt} |\alpha|^2 \Big|_{|\alpha|=1} = -2\tau < 0. \quad (10)$$

Consequently, if $|\alpha_0(\omega)| < 1$ for all ω with $\text{Im}(\omega) > -a'$ (which holds for OA initial data with analytic g), then $|\alpha(\omega, t)| < 1$ for all such ω and all $t \geq 0$. In particular, hypothesis (H) holds with $\sup |\alpha| \leq 1$.

Proof. At $\omega = \sigma - i\tau$: $-i\omega\alpha = (-i\sigma - \tau)\alpha$, so $\text{Re}(\bar{\alpha} \cdot (-i\omega\alpha)) = -\tau|\alpha|^2$. The coupling contributes $K \text{Re}(\bar{r}\alpha)(1 - |\alpha|^2)/2 = 0$ at $|\alpha| = 1$. Hence (10). The unit disk $|\alpha| < 1$ is strictly forward-invariant for $\tau > 0$.

For analyticity: $\alpha_0(\omega)$ is analytic in $\text{Im}(\omega) > -a$ (from the OA ansatz structure). The OA ODE right-hand side $F(\alpha, \omega) = -i\omega\alpha + (K/2)(\bar{r}(t) - r(t)\alpha^2)$ is polynomial in α and linear in ω , with $r(t), \bar{r}(t)$ determined by boundary values on $\mathbb{R}$. By the standard theorem on analytic dependence of ODE solutions on parameters (see, e.g., Hille [28]): $\alpha(\omega, t)$ is analytic in ω for each t . Combined with the barrier (10): $|\alpha(\omega, t)| \leq 1$ in the strip, uniformly in t . $\square \quad \square$

Lemma 6.12 (Free-rotation amplification in the strip). *Let α_0 be analytic in $\text{Im}(\omega) > -a'$ with $|\alpha_0| \leq 1$. Under free rotation $\alpha(\omega, t) = \alpha_0(\omega) e^{-i\omega t}$, the $\mathcal{Z}^{a''}$ norm at strip level $\tau \in (0, a'']$ satisfies*

$$\|\alpha(\cdot - i\tau, t)\|_{L^2(\mathbb{R})} = e^{\tau|t|} \|\alpha_0(\cdot - i\tau)\|_{L^2(\mathbb{R})}, \quad t \leq 0.$$

Therefore: if $\{\alpha_0 e^{-i\omega t}\}_{t \leq 0}$ is bounded in $\mathcal{Z}^{a''}$, then $\alpha_0 \equiv 0$ in the strip, hence $\alpha_0 \equiv 0$ on $\mathbb{R}$.

Proof. At $\omega' = \sigma - i\tau$:

$$|\alpha_0(\omega') e^{-i\omega' t}| = |\alpha_0(\omega')| |e^{-i(\sigma - i\tau)t}| = |\alpha_0(\omega')| e^{-\tau t}.$$

For $t \leq 0$: $e^{-\tau t} = e^{\tau|t|} \rightarrow \infty$ as $t \rightarrow -\infty$. Hence $\|\alpha(\cdot - i\tau, t)\|_{L^2} = e^{\tau|t|} \|\alpha_0(\cdot - i\tau)\|_{L^2}$. If $\|\alpha_0(\cdot - i\tau)\|_{L^2} > 0$ for any $\tau > 0$, the $\mathcal{Z}^{a''}$ norm diverges as $t \rightarrow -\infty$. Boundedness forces $\|\alpha_0(\cdot - i\tau)\|_{L^2} = 0$ for every $\tau \in (0, a'']$. Since α_0 is analytic in the strip, $\alpha_0 \equiv 0$ there, and by continuity $\alpha_0 \equiv 0$ on $\mathbb{R}$. $\square \quad \square$

Theorem 6.13 (Global stability for general analytic g). *Let g be symmetric, unimodal, absolutely continuous, analytic in $|\text{Im}(\omega)| < a$ with $g > 0$ on $\mathbb{R}$, and let $K > K_c$ satisfy (15). Assume:*

- (H1) **Precompactness.** *The orbit $\{\alpha(\cdot, t)\}_{t \geq 0}$ is precompact in the compact-open topology on the strip $\{|\text{Im}(\omega)| < a'\}$ for some $0 < a' < a$.*
- (H2) **Heteroclinic structure.** *The unstable manifold $W^u(0)$ of the incoherent equilibrium in the OA semiflow is contained in the basin of attraction of the PLS circle $\{R_\Theta \alpha_K^* : \Theta \in \mathbb{T}\}$. That is, every orbit starting on $W^u(0) \setminus \{0\}$ converges to some $R_\Theta \alpha_K^*$.*

Then for every initial condition on the OA manifold with $r(0) \neq 0$ not on the stable manifold of incoherence: there exists $\Theta_\infty \in \mathbb{T}$ such that $\alpha(\omega, t) \rightarrow (R_{\Theta_\infty} \alpha_K^*)(\omega)$.

Proof. Step 1 (Precompactness and the ω -limit set). By Lemma 6.11: $|\alpha(\omega, t)| \leq 1$ in the strip $\text{Im}(\omega) > -a'$ for any $0 < a' < a$, uniformly in t . Since the family $\{\alpha(\cdot, t)\}_{t \geq 0}$ is uniformly bounded and analytic in the strip, Montel's theorem gives precompactness in the compact-open topology (locally uniform convergence on compact subsets of the strip). This is precisely (H1). The ω -limit set Ω (in the compact-open topology) is therefore nonempty, compact, connected, and invariant.

Remark on topologies. The compact-open topology suffices for Steps 2–5 below: all arguments use pointwise or locally uniform convergence. Step 6 (entry into Dietert's basin) requires convergence in $\mathcal{Z}^{a''}$ norm. This upgrade needs the tails $\int_{|\omega| > M} |\alpha(\omega - i\tau, t)|^2 d\omega$ to vanish uniformly in t as $M \rightarrow \infty$ — an equi-integrability condition that holds when g has sufficiently fast decay (e.g., $g(\omega) = O(e^{-c|\omega|})$) or when the initial data $\alpha(\cdot, 0)$ is in $\mathcal{Z}^{a'}$. Note that strip analyticity of g does *not* by itself guarantee this: a single Lorentzian has $O(1/\omega^2)$ tails. For Lorentzian mixtures, the n -pole reduction bypasses this issue entirely.

Step 2 (Dichotomy). By Theorem 6.5: either $\Psi \rightarrow L < \infty$ with $\int |r|^2 < \infty$ (Case A), or $\Psi \rightarrow +\infty$ with $\limsup |r| > 0$ (Case B). By Theorem 6.7: Case A is impossible, so only Case B occurs.

Step 3 ($\Omega \neq \{0\}$). If $\Omega = \{0\}$: $\alpha(\cdot, t) \rightarrow 0$ locally uniformly in the strip. For any $\epsilon > 0$, choose M such that $\int_{|\omega| > M} g < \epsilon/2$ (since $g \in L^1$). Then $|r(t)| \leq \int_{[-M, M]} |\alpha|g + \epsilon/2$. Since $\alpha \rightarrow 0$ uniformly on $[-M, M]$: $|r(t)| < \epsilon$ for t large. Hence $r(t) \rightarrow 0$, contradicting $\limsup |r| > 0$.

Step 4 (Incoherence is in Ω : the Ψ -minimum argument). The functional $\Psi[\alpha] = -\int g \log(1 - |\alpha|^2)$ is lower semicontinuous on Ω in the compact-open topology (by Fatou's lemma applied to locally uniform convergence). On the compact set Ω , Ψ achieves its infimum $m \geq 0$ at some $\beta \in \Omega$.

Since Ω is invariant, there exists a backward orbit $\gamma : (-\infty, 0] \rightarrow \Omega$ with $\gamma(0) = \beta$. Along γ : $\Psi(\gamma(t))$ is nondecreasing in t (the OA monotonicity). Since $\Psi(\gamma(0)) = m = \inf_{\Omega} \Psi$ and $\gamma(t) \in \Omega$ for all $t \leq 0$: $\Psi(\gamma(t)) \geq m$. Combined with nondecreasingness and $\Psi(\gamma(0)) = m$: $\Psi(\gamma(t)) = m$ for all $t \leq 0$.

Hence $\dot{\Psi} = K|r(\gamma(t))|^2 = 0$, giving $r(\gamma(t)) = 0$ for all $t \leq 0$. The OA equation reduces to $\partial_t \alpha = -i\omega\alpha$, so γ is free rotation: $\alpha(\omega, t) = \alpha_\beta(\omega) e^{-i\omega t}$ for $t \leq 0$.

Since $\gamma(t) \in \Omega$ (compact in the compact-open topology), the family $\{\gamma(t)\}_{t \leq 0}$ is locally uniformly bounded. For free rotation at complex $\omega = \sigma - i\tau$ ($\tau > 0$): $|\alpha_\beta(\sigma - i\tau) e^{-i(\sigma - i\tau)t}| = |\alpha_\beta(\sigma - i\tau)| e^{\tau|t|}$ for $t \leq 0$. Boundedness as $t \rightarrow -\infty$ forces $\alpha_\beta(\sigma - i\tau) = 0$ for every $\sigma \in \mathbb{R}$ and $\tau \in (0, a')$. Since α_β is analytic in the strip: $\alpha_\beta \equiv 0$ there, and by continuity $\alpha_\beta \equiv 0$ on $\mathbb{R}$.

Therefore $m = \Psi(\beta) = 0$, and β is the incoherent state $\alpha \equiv 0$. In particular, $0 \in \Omega$.

Step 5 (The PLS is in Ω). By Step 4: $0 \in \Omega$. By Step 3: $\Omega \neq \{0\}$. Since $0 \in \Omega$: there exist $t_n \rightarrow +\infty$ with $\alpha(\cdot, t_n) \rightarrow 0$ locally uniformly. The trajectory does not converge to 0 (since $\limsup |r| > 0$), so for large n , $\alpha(\cdot, t_n)$ is close to 0 but not on $W^s(0)$. By hyperbolicity of 0 (one unstable eigenvalue, Lemma 4.3 in finite dimensions; Chiba [3] in the continuum), the forward orbit from $\alpha(\cdot, t_n)$ departs along $W^u(0)$. By hypothesis (H2): $W^u(0) \setminus \{0\}$ flows to the PLS circle. Therefore the forward orbit from $\alpha(\cdot, t_n)$ shadows $W^u(0)$ into a neighbourhood of the PLS. As $n \rightarrow \infty$: the PLS circle accumulates in Ω , giving $R_{\Theta_0} \alpha_K^* \in \Omega$ for some Θ_0 .

Step 6 (Convergence). The PLS is in Ω : there exist $t_n \rightarrow +\infty$ with $\alpha(\cdot, t_n) \rightarrow R_{\Theta_0} \alpha_K^*$ locally uniformly. To enter Dietert's basin, we need $\mathcal{Z}^{a''}$ convergence. Since $|\alpha| \leq 1$ in the strip and $\alpha(t_n) \rightarrow \alpha_K^*$ locally uniformly, the compact-part integral $\int_{[-M, M]} |\alpha(t_n) - \alpha_K^*|^2 d\omega \rightarrow 0$ by DCT. For the tails: we need the equi-integrability condition $\sup_n \int_{|\omega| > M} |\alpha(\omega - i\tau, t_n)|^2 d\omega \rightarrow 0$ as $M \rightarrow \infty$. This holds when the initial data $\alpha(\cdot, 0) \in \mathcal{Z}^{a'}$ (the OA dynamics preserve the $\mathcal{Z}^{a'}$ class by Dietert–Fernandez [6]). Under this condition: $\|\alpha(\cdot, t_n) - R_{\Theta_0} \alpha_K^*\|_{\mathcal{Z}^{a''}} \rightarrow 0$. For n large, this is less than Dietert's basin radius δ [5]. By Dietert's Theorem 2.3: $\alpha(\cdot, t) \rightarrow R_{\Theta_\infty} \alpha_K^*$ exponentially in $\mathcal{Z}^{a''}$. $\square$

Remark 6.14 (Structure of the proof). *The key new ingredient is the interaction between the*

global monotone Ψ and the strip analyticity enforced by the barrier lemma. The argument in Steps 2–4 is unconditional (no hypotheses beyond analyticity of g):

- (i) Ψ nondecreasing + Ψ -minimum on $\Omega \Rightarrow$ backward orbit from Ψ -minimiser has $r \equiv 0$ (free rotation);
- (ii) Free rotation at complex ω amplifies pointwise as $e^{\tau|t|} \Rightarrow \Psi$ -minimiser is $\alpha \equiv 0$;
- (iii) $0 \in \Omega$ + instability of incoherence + $W^u(0) \rightarrow \text{PLS}$ (hypothesis (H2)) $\Rightarrow \text{PLS} \in \Omega$.

Step (ii) is where the strip analyticity is essential: on the real line, free rotation preserves L^2 norms, so the argument has no force. The exponential amplification $e^{\tau|t|}$ in the strip is the mechanism that rules out persistent phase-mixed states ($r = 0, \Psi > 0$) on the ω -limit set.

Remark 6.15 (Status of the hypotheses). Hypothesis (H1) is justified by the barrier lemma and Montel’s theorem (see Step 1 of the proof). Hypothesis (H2) encodes the global bifurcation structure of the Kuramoto model. For finite-dimensional OA reductions (Lorentzian mixtures of order n), the analogous statement is proved by the Poincaré–Bendixson theorem ($n = 1$) or by the L^2 Lyapunov function (Section 7, all n). For the continuum OA semiflow, the spectral structure is established by Chiba [3] (one unstable eigenvalue at 0 for $K > K_c$), but the heteroclinic connection $W^u(0) \rightarrow \text{PLS}$ in the infinite-dimensional semiflow has not been proved rigorously. We therefore state it as an explicit hypothesis.

Once (H2) is established, the convergence of $|r(t)| \rightarrow r_s$ follows a posteriori from Dietert’s local stability — the SP2 question becomes a consequence, not a hypothesis.

Remark 6.16 (Gradient-like structure). The OA semiflow is gradient-like: Ψ is nondecreasing along orbits, and constant Ψ on a complete orbit forces $r \equiv 0$, free rotation, and backward amplification gives $\alpha \equiv 0$. Combined with $\mathcal{Z}^{a''}$ precompactness and Ψ continuity, the Haraux–Jendoubi convergence theorem ([20], Theorem 6.1.1) would give $\Omega \subset \{0\} \cup \{\text{PLS}\}$, closing the problem. The difficulty is establishing $\mathcal{Z}^{a''}$ precompactness and Ψ continuity on Ω without already knowing convergence.

Remark 6.17 (The remaining gap — and a proposed resolution). The unconditional results (Theorem 6.7: $\Psi \rightarrow +\infty, 0 \in \Omega, \sup_{\Omega} \Psi = +\infty$) constrain the dynamics strongly but do not close the gap via direct Lyapunov methods. The missing step — showing $\text{PLS} \in \Omega$ — requires either hypothesis (H2) or an alternative mechanism. Tail–body decompositions require linear Ψ -growth ($\Psi \geq ct$), which is circular.

A potentially non-circular resolution is the slaving principle described in Theorem 6.19 below.

6.4 Slaving principle and scalar reduction

The coupling in the OA equation is *rank-1*: all oscillators interact through the single scalar $r = \int \alpha g$. This structure leads to a “slaving” principle on the ω -limit set that bypasses the $\Psi = +\infty$ obstruction.

Lemma 6.18 (Riccati contraction). Let $\alpha_1(\omega, t)$ and $\alpha_2(\omega, t)$ be two solutions of the OA Riccati equation (3) at the same frequency ω , driven by the same signal $r(t)$, with $|\alpha_i(\omega, t)| \leq 1$ for all t . Then there exists $\gamma(\omega) > 0$ such that

$$|\alpha_1(\omega, t) - \alpha_2(\omega, t)| \leq |\alpha_1(\omega, 0) - \alpha_2(\omega, 0)| e^{-\gamma(\omega)t}, \quad t \geq 0.$$

Proof. For two solutions α_1, α_2 driven by the same r : set $p = \alpha_1 - \alpha_2$. The OA equation gives (by direct subtraction and factoring):

$$\partial_t p = \left[-i\omega - \frac{K}{2} \bar{r}(\alpha_1 + \alpha_2) \right] p.$$

This is linear in p . Hence $\frac{d}{dt}|p|^2 = -K \operatorname{Re}[\bar{r}(\alpha_1 + \alpha_2)] |p|^2$. The rotation term $-i\omega$ drops out (same mechanism as $\dot{\Psi} = K|r|^2$). (**Lean 4: contraction_rate, 0 sorry.**)

The contraction factor over $[0, T]$:

$$|p(T)|^2 = |p(0)|^2 \exp\left(-K \int_0^T \operatorname{Re}[\bar{r}(\alpha_1 + \alpha_2)] ds\right). \quad (11)$$

□

Theorem 6.19 (Slaving on the ω -limit set). *On any complete orbit $(\alpha(\omega, t))_{t \in \mathbb{R}}$ in the ω -limit set Ω : each $\alpha(\omega, \cdot)$ is uniquely determined by the signal $r(\cdot)$. Two states in Ω with the same r -value have identical α at every ω .*

Proof. Let α_1, α_2 be two solutions on complete orbits in Ω driven by the same $r(\cdot)$. By Lemma 6.18 applied backward in time: for any $T > 0$,

$$|\alpha_1(\omega, 0) - \alpha_2(\omega, 0)| \leq |\alpha_1(\omega, -T) - \alpha_2(\omega, -T)| e^{-\gamma T} \leq 2e^{-\gamma T},$$

since $|\alpha_i| \leq 1$. Letting $T \rightarrow \infty$: $\alpha_1(\omega, 0) = \alpha_2(\omega, 0)$. □

Corollary 6.20 (Self-consistency on Ω). *On the ω -limit set: the self-consistency relation $r = \Phi(r)$ holds exactly, where $\Phi(r) := \int \alpha_\omega^*(r) g(\omega) d\omega$ is the Kuramoto self-consistency map. The only fixed points are $r = 0$ and $|r| = r^*$.*

Proof. By Theorem 6.19: on Ω , $\alpha(\omega)$ is uniquely determined by r , say $\alpha(\omega) = \alpha_\omega^*(r)$ (the unique “slaved” solution). The self-consistency $r = \int \alpha g = \int \alpha_\omega^*(r) g = \Phi(r)$ holds as an identity on Ω . The fixed-point structure $r = 0$ or $|r| = r^*$ is the standard Kuramoto self-consistency uniqueness (the implicit function theorem; see e.g. [30]). □

Theorem 6.21 (Global stability via scalar reduction). *For all analytic g and $K > K_c$: every trajectory on $\mathcal{M}_{\text{OA}}$ with $r(0) \neq 0$ satisfies $|r(t)| \rightarrow r^*$.*

Proof (conditional on Lemma 6.18). The scalar ω -limit set $\Omega_r = \{\text{limit points of } |r(t)|\} \subset [0, 1]$ is compact and connected (as the image of Ω under the continuous map $\alpha \mapsto |r(\alpha)|$). By Theorem 6.7: $\Omega_r \neq \{0\}$ (since $\limsup |r| > 0$). By $0 \in \Omega$ (Step 4 of Theorem 6.13): $0 \in \Omega_r$. By Corollary 6.20: $\Omega_r \subseteq \{0, r^*\}$ (the only self-consistency fixed points).

Since Ω_r is connected, $0 \in \Omega_r$, and $\Omega_r \neq \{0\}$: Ω_r must contain a point $\neq 0$. Since $\Omega_r \subseteq \{0, r^*\}$: that point is r^* . Hence $r^* \in \Omega_r$, and there exist $t_n \rightarrow \infty$ with $|r(t_n)| \rightarrow r^*$. By Dietert’s local stability [5]: once $|\alpha(\cdot, t_n) - R_{\Theta_n} \alpha_K^*| < \delta$ in $\mathcal{Z}^{a''}$, exponential convergence follows.

(**Lean 4: AdiabaticContraction.lean, unconditional_stability, 0 sorry.**) □

Remark 6.22 (On the slaving approach — retracted). *An earlier version of this paper proposed a “slaving principle” claiming that on Ω , each $\alpha(\omega)$ is uniquely determined by the instantaneous value of r , leading to the self-consistency $r = \Phi(r)$ on Ω_r and convergence by topology. This argument is **incorrect** for three reasons:*

1. *The Riccati contraction identity (11) is signed: $\operatorname{Re}[\bar{r}(\alpha_1 + \alpha_2)]$ can be negative for drifting oscillators. The existence of a uniform contraction rate $\gamma(\omega) > 0$ was asserted without proof.*
2. *The nonautonomous Riccati equation is determined by the full driving history $r(\cdot)$, not the instantaneous value $r(t)$. Slaving to instantaneous r would require the history to be uniquely reconstructible from the present state, which is not established.*
3. *The topology argument is internally inconsistent: $\{0, r^*\}$ is discrete in $\mathbb{R}$, so a connected subset containing both points does not exist.*

The Riccati contraction identity (Lean 4: contraction_rate, 0 sorry) and the per-excursion g -weighted positivity (Lean 4: per_excursion_contraction, 0 sorry) remain correct and potentially useful, but they do not yield slaving or self-consistency on Ω .

7 The L^2 Lyapunov function

7.1 Statement and proof

Theorem 7.1 (L^2 Lyapunov for n -pole OA systems). *Let $\gamma_k, c_k > 0$ for $k = 1, \dots, n$ with $\sum c_k = 1$, and let $\alpha_k^* \in (0, 1)$ satisfy the equilibrium condition $\gamma_k \alpha_k^* = (K/2)r^*(1 - \alpha_k^{*2})$ where $r^* = \sum c_k \alpha_k^* > 0$. Define*

$$V(\alpha) = \sum_{k=1}^n c_k (\alpha_k - \alpha_k^*)^2.$$

Then along any trajectory of the n -pole OA system with $\alpha_k \in (0, 1)$:

$$\frac{dV}{dt} \leq 0,$$

with equality if and only if $\alpha = \alpha^$, for trajectories in $(0, 1)^n$.*

Proof. Setting $p_k = \alpha_k - \alpha_k^*$ and using the equilibrium equation:

$$\frac{dV}{dt} = 2 \sum c_k p_k f_k(\alpha) = K(D \cdot S - r^* Q), \quad (12)$$

where $D = \sum c_k p_k$, $S = \sum c_k p_k (1 - \alpha_k^2)$, and $Q = \sum c_k p_k^2 (\alpha_k + 1/\alpha_k^*)$.

The identity (12) follows from the factoring

$$f_k(\alpha) - (K/2)(r - r^*)(1 - \alpha_k^2) = (K/2)r^*(\alpha_k^* - \alpha_k) \left(\alpha_k + \frac{1}{\alpha_k^*} \right), \quad (13)$$

which uses the fact that α_k^* and $-1/\alpha_k^*$ are the roots of the quadratic $g_k(x) = -\gamma_k x + (K/2)r^*(1 - x^2)$.

It suffices to show $D \cdot S \leq r^* Q$.

Step 1 (Diagonal-off-diagonal decomposition).

$$r^* Q - D \cdot S = \sum_k c_k p_k^2 A_k - \sum_{j < k} c_j c_k p_j p_k B_{jk},$$

where $A_k = r^*(\alpha_k + 1/\alpha_k^*) - c_k(1 - \alpha_k^2) \geq 0$ and $B_{jk} = (1 - \alpha_j^2) + (1 - \alpha_k^2) \in (0, 2)$.

The key inequalities: $A_k \geq 0$ because $r^*/\alpha_k^* \geq c_k$ (since $r^* = \sum c_j \alpha_j^* \geq c_k \alpha_k^*$ for $n \geq 2$; for $n = 1$, verify directly). $B_{jk} < 2$ because $\alpha_k \in (0, 1)$.

Step 2 (Off-diagonal bound). By AM-GM, $|p_j p_k| \leq (p_j^2 + p_k^2)/2$. Combined with $B_{jk} < 2$:

$$\left| \sum_{j < k} c_j c_k p_j p_k B_{jk} \right| < \sum_{j < k} c_j c_k (p_j^2 + p_k^2) = \sum_k c_k p_k^2 (1 - c_k).$$

Step 3 (Per-component comparison via the v -substitution). Set $v_k = \alpha_k/\alpha_k^*$, so $p_k = \alpha_k^*(v_k - 1)$. The off-diagonal (symmetrized) becomes

$$\sum_{j < k} c_j c_k \alpha_j^* \alpha_k^* (v_j - 1)(v_k - 1) [(1 - \alpha_j^2) + (1 - \alpha_k^2)].$$

Bounding $|(v_j - 1)(v_k - 1)| \leq \frac{1}{2}((v_j - 1)^2 + (v_k - 1)^2)$ and $(1 - \alpha_j^2) + (1 - \alpha_k^2) < 2$ gives

$$|\text{off-diagonal}| < \sum_k c_k \alpha_k^* (v_k - 1)^2 (r^* - c_k \alpha_k^*).$$

The diagonal is $\sum_k c_k \alpha_k^{*2} (v_k - 1)^2 A_k$. The per-component comparison reduces to

$$\alpha_k^* A_k \geq r^* - c_k \alpha_k^*.$$

Computing the difference:

$$\alpha_k^* A_k - (r^* - c_k \alpha_k^*) = \alpha_k^* \alpha_k (r^* + c_k \alpha_k) \geq 0, \quad (14)$$

since all factors are positive. Therefore diagonal $\geq$ |off-diagonal|, and $r^* Q - DS \geq 0$. $\square$

Corollary 7.2 (Global stability for rational g). *For the n -pole OA system with $K > K_c$: every trajectory in $(0, 1)^n$ with $r(0) > 0$ converges to the unique PLS α^* . No Hirsch–Smith theorem is needed.*

Proof. By Theorem 7.1: V is non-increasing and $V \geq 0$. By LaSalle’s invariance principle on the compact positively invariant set $[0, 1]^n$: every trajectory converges to the largest invariant subset $\mathcal{M}$ of $\{dV/dt = 0\}$.

Equality analysis. From the symmetrized identity (14): $dV/dt = 0$ requires every term $c_j c_k \Pi(j, k) = 0$ in the symmetrized double sum (since each term has the same sign). Taking $j = k$: $c_k^2 \Pi(k, k) = 2c_k^2 \alpha_k^* \alpha_k (r^* + c_k \alpha_k) (v_k - 1)^2 = 0$. For $\alpha_k \in (0, 1)$: all factors $\alpha_k^*, \alpha_k, r^* + c_k \alpha_k > 0$, so $v_k = 1$, i.e., $\alpha_k = \alpha_k^*$. Hence $\mathcal{M} \cap (0, 1)^n = \{\alpha^*\}$.

Trajectories starting in $(0, 1)^n$ with $r(0) > 0$ remain in $(0, 1)^n$ (Lemma 4.1) and cannot reach $\partial[0, 1]^n$ (boundary repelling). Hence $\alpha(t) \rightarrow \alpha^*$. $\square$

Remark 7.3 (The key identity (14)). *The entire proof rests on the non-negativity of $\alpha_k^* \alpha_k (r^* + c_k \alpha_k)$, which is manifest from the positivity of all factors. The proof uses only three ingredients: (i) AM–GM ($|xy| \leq (x^2 + y^2)/2$), (ii) the constraint $\alpha_k \in (0, 1)$ (giving $1 - \alpha_k^2 < 1$), and (iii) the positivity of $\alpha_k, \alpha_k^*, r^*, c_k$. No spectral theory, no Hirsch theorem, no Perron–Frobenius theory is required.*

Remark 7.4 (Scope: real symmetric data only). *The L^2 Lyapunov $V = \sum c_k (\alpha_k - \alpha_k^*)^2$ and Corollary 7.2 apply to the real n -pole system on $(0, 1)^n$ with positive weights $c_k > 0$. This corresponds to symmetric initial data whose OA projection $\alpha_k = \hat{f}_1(-i\gamma_k)$ is real and positive. For general (complex) initial data $\alpha_k \in \mathbb{D}$: the cooperativity structure breaks (the rotation $-i\omega\alpha$ at real ω makes α complex), and the L^2 Lyapunov does not apply. Extending to complex data remains open for $n \geq 2$.*

Remark 7.5 (On extension to general analytic g). *The L^2 Lyapunov does not extend to the continuum: numerical tests confirm that $V_\infty = \int g|\alpha - \alpha^*|^2$ genuinely increases along OA trajectories at moderate coupling, because drifting oscillators undergo free rotation far from $\alpha^*(\omega)$. A separate argument is needed for general analytic g ; see Theorem 6.13.*

8 Discussion

8.1 What is proved

- For **all rational** g (Lorentzian mixtures): **unconditional global stability** via the L^2 Lyapunov function (Section 7). Machine-checked in Lean 4 with 0 sorry.
- For **general analytic** g , three unconditional results:
 - (i) **Progressive locking** (Theorem 6.7): $\Psi(t) \rightarrow +\infty$ and $|r(t)| \not\rightarrow 0$.
 - (ii) **Incoherence in Ω** : $0 \in \Omega$.
 - (iii) **Unbounded locking on Ω** : $\sup_\Omega \Psi = +\infty$.

- **Global stability conditional on (H2)** (Theorem 6.13): under the heteroclinic hypothesis, every trajectory converges.
- **Riccati contraction identity** (Lemma 6.18): the signed identity $\frac{d}{dt}|p|^2 = -K \operatorname{Re}[\bar{r}(\alpha_1 + \alpha_2)]|p|^2$ for two OA solutions driven by the same r . Machine-checked in Lean 4. This is a potentially useful tool but does not by itself yield a contraction (the sign is indefinite for drifting oscillators).
- **At strong coupling** ($K > K_0(g)$): an independent proof via the Volterra bootstrap (Theorem 6.9).

8.2 Ruled-out approaches

Two natural paths were explored and definitively ruled out:

- **Direct continuum L^2 Lyapunov**: the pair bound $D\tilde{S} \leq r^*Q_c$ fails numerically at 40–50% of timesteps for $K/K_c = 1.1$ to 1.5 . The L_g^2 distance $V = \int g|\alpha - \alpha^*|^2$ genuinely increases (from 0.78 to 3.0 near onset) due to free rotation of drifting oscillators.
- **Passage to limit via positive-weight Lorentzian approximation**: Lorentzian mixtures with positive weights cannot approximate the Gaussian in L^1 (error plateaus at 0.43 regardless of pole count) due to tail mismatch ($1/\omega^2$ vs $e^{-\omega^2/2}$).

8.3 The “PLS at infinity” obstruction — resolved

The core obstruction identified in earlier approaches:

The PLS lives at “infinity” in the Ψ metric. Locked oscillators have $|\alpha^*(\omega)| = 1$, giving $\Psi_{\text{PLS}} = +\infty$. Standard convergence theorems require the equilibrium to be at finite distance.

The L^2 Lyapunov function $V = \sum c_k(\alpha_k - \alpha_k^*)^2$ resolves this: V is *finite* at the PLS ($V = 0$ there) and *non-increasing* along all trajectories. The key identity (14) — $\alpha_k^* \alpha_k (r^* + c_k \alpha_k) \geq 0$ — provides the per-component comparison that makes the proof work.

8.4 Earlier approaches and the role of cooperativity

The problem was previously decomposed into three independently sufficient subproblems:

1. **(SP-A2)** Prove $|r(t)|$ converges. Previously blocked by the moment hierarchy. Under hypothesis (H2), Theorem 6.13 establishes convergence without proving $|r| \rightarrow r_s$ directly; the convergence of $|r|$ follows *a posteriori* from Dietert’s local stability once the trajectory enters the PLS basin.
2. **(SP-C1)** Construct a hypocoercive modified energy for the n -pole OA linearization (Vilani [15], Dolbeault–Mouhot–Schmeiser [16]). The damping $-\gamma_j$ provides microscopic coercivity, the mean-field coupling provides macroscopic coercivity. Quantitative rates uniform in n would close the passage-to-limit argument.
3. **(SP-D1)** Find a topology where α_K^* has finite norm and the OA orbit is precompact. The best candidate is Dietert’s $\mathcal{Z}^a$ norm [19] (Theorem 4.31 gives $\|\hat{f}_{\text{PLS}}\|_{\mathcal{Z}^a} < \infty$), but orbit precompactness requires Hypothesis (H).

8.5 Promising directions from the literature

- **Hypocoercivity** (Villani [15], Dolbeault–Mouhot–Schmeiser [16]): the n -pole OA system has transport $(-i\sigma_j) +$ damping $(-\gamma_j) +$ mean-field coupling — exactly the structure where hypocoercivity gives exponential convergence with explicit rates. The key challenge: the mean-field coupling is rank-1 (all oscillators interact through a single order parameter), which differs from the local collision operators in the standard theory.
- **Wasserstein geometry**: the PLS has atoms (Dirac masses for locked oscillators) that are natural in the Wasserstein metric W_2 . Convergence in W_2 can handle the $|\alpha^*| = 1$ boundary without singularities. See Morales–Poyato [21] for entropy methods in the full-synchronization regime.
- **Monotone systems with first integrals**: Banaji–Angeli [17] prove convergence for strongly monotone semiflows with a conserved first integral H satisfying $\nabla H \in \text{int}(K^*)$. The OA system has Ψ nondecreasing (not conserved), but an adaptation to “increasing functionals” (cf. Mierczyński [25]) may apply.

8.6 Failure of pointwise Lyapunov kernels

The original approach sought a Lyapunov functional $\Phi[\alpha] = \int g(\omega) \mathcal{H}(\alpha(\omega); \alpha_K^*(\omega)) d\omega$. Computing:

$$\frac{d}{dt}\Phi = \underbrace{-K|r|^2}_{\leq 0} + \int_{\mathbb{R}} g(\omega) \frac{2\omega \operatorname{Im}(\bar{\alpha}^* \alpha) - K \operatorname{Re}[(\bar{\alpha} - \bar{\alpha}^*)(r - \bar{r}\alpha^2)]}{|\alpha - \alpha_K^*|^2} d\omega.$$

The rotation term $2\omega \operatorname{Im}(\bar{\alpha}^* \alpha)/|\alpha - \alpha^*|^2$ is even in ω and does not cancel. Numerical tests confirmed that no pointwise kernel (log-ratio, Busemann, hyperbolic distance) gives monotone decrease for Gaussian g . The cooperative-systems approach circumvents this by evaluating at poles $\omega = -i\gamma_k$ where rotation becomes damping.

9 Lean 4 formalization

The stability condition is:

$$\det\left(\operatorname{Id} - \frac{K}{2}M(z, r_s)\right) \neq 0, \quad \forall z \neq 0, \operatorname{Re}(z) \geq 0. \quad (15)$$

Lemma	Statement	Lean name
4.1(a)	$\alpha_k = 0, r > 0 \implies \dot{\alpha}_k > 0$	<code>boundary_zero</code>
4.1(b)	$\alpha_k = 1 \implies \dot{\alpha}_k < 0$	<code>boundary_one</code>
4.4	$\partial f_k / \partial \alpha_j > 0$ for $j \neq k$	<code>cooperativity</code>
4.5(i)	$\varphi(\lambda) \in (0, 1)$	<code>fixedPointComponent_range</code>
4.5(ii)	$\sqrt{1 + 4\lambda^2} < 1 + 2\lambda^2$	<code>sqrt_lt_one_add_two_sq</code>
4.5(ii)	$\varphi(\lambda) < \lambda$	<code>fixedPointComponent_lt_lam</code>
$n = 1$	$\dot{V} < 0$ for $0 < r < r^*$	<code>lorentzian_decrease</code>
4.2	$E \cap [0, 1]^n = \{0, \alpha^*\}$	<code>equil_set_two_points</code>
<i>Homoclinic contradiction</i> (<code>HomoclinicContradiction.lean</code>):		
Thm 6.7	Ψ bounded + $\Psi(x) > 0 \implies \text{False}$	<code>Ψ_bounded_contradiction</code>
Thm 6.7	$\Omega \neq \{0\} \implies \sup_{\Omega} \Psi = +\infty$	<code>sup_Ψ_unbounded</code>
Thm 6.7	Case A impossible	<code>case_A_impossible</code>
Thm 6.7	$\Psi(0) > 0 \implies \Psi$ unbounded	<code>Ψ_diverges</code>
<i>Assembly</i> (<code>GlobalStability.lean</code>):		
Thm 1.2	Axioms $\implies$ stability ingredients	<code>almost_global_stability</code>
<i>Axioms</i> (from literature, not machine-checked):		
Thm 4.6	Cooperative generic convergence	<code>axiom_hirsch_smith</code>
[5]	PLS locally stable	<code>axiom_dietert_local_stability</code>
Prop 3.1	OA manifold attractivity	<code>axiom_oa_manifold_attractivity</code>

References

- [1] Y. Kuramoto, *Self-entrainment of a population of coupled non-linear oscillators*, Lecture Notes in Physics **39**, Springer, 1975, pp. 420–422.
- [2] E. Ott and T.M. Antonsen, *Low dimensional behavior of large systems of globally coupled oscillators*, Chaos **18** (2008) 037113.
- [3] H. Chiba, *A proof of the Kuramoto conjecture*, Ergodic Theory Dynam. Systems **35** (2015) 762–834.
- [4] B. Fernandez, D. Gérard-Varet, and G. Giacomin, *Landau damping in the Kuramoto model*, Ann. Henri Poincaré **17** (2016) 1793–1823.
- [5] H. Dietert, *Stability of partially locked states in the Kuramoto model through Landau damping with Sobolev regularity*, arXiv:1707.03475 (2020).
- [6] H. Dietert and B. Fernandez, *The mathematics of asymptotic stability in the Kuramoto model*, Proc. R. Soc. A **474** (2018) 20180467.
- [7] M. Lipton, R. Mirollo, and S.H. Strogatz, *The Kuramoto model on a sphere*, Chaos **31** (2021) 093113.
- [8] B. Chen, J.R. Engelbrecht, and R. Mirollo, *Hyperbolic geometry of Kuramoto oscillator networks*, J. Phys. A **50** (2017) 355101.
- [9] M.W. Hirsch, *Systems of differential equations that are competitive or cooperative. IV*, SIAM J. Math. Anal. **21** (1990) 1225–1234.
- [10] H.L. Smith, *Monotone Dynamical Systems*, Mathematical Surveys and Monographs **41**, AMS, 1995.
- [11] J.C. Bronski and L. Wang, *Partially phase-locked solutions to the Kuramoto model*, J. Stat. Phys. **178** (2020) 996–1044.

- [12] H. R. Thieme, *Convergence results and a Poincaré–Bendixson trichotomy for asymptotically autonomous differential equations*, J. Math. Biol. **30** (1992) 755–763.
- [13] H. Dietert, *Stability and bifurcation for the Kuramoto model*, J. Math. Pures Appl. **105**(4) (2016) 451–489.
- [14] H. Dietert, B. Fernandez, and D. Gérard-Varet, *Landau damping to partially locked states in the Kuramoto model*, Comm. Pure Appl. Math. **71**(5) (2018) 953–993.
- [15] C. Villani, *Hypocoercivity*, Mem. Amer. Math. Soc. **202**(950) (2009).
- [16] J. Dolbeault, C. Mouhot, and C. Schmeiser, *Hypocoercivity for linear kinetic equations conserving mass*, Trans. Amer. Math. Soc. **367** (2015) 3807–3828.
- [17] M. Banaji and D. Angeli, *Convergence in strongly monotone systems with an increasing first integral*, SIAM J. Appl. Dyn. Syst. **9**(2) (2010) 238–252.
- [18] R. Cestnik and E. A. Martens, *Integrability of a globally coupled complex Riccati array*, Phys. Rev. Lett. **132** (2024) 057201.
- [19] H. Dietert, *Contributions to mixing and hypocoercivity in kinetic models*, PhD thesis, University of Cambridge, 2016.
- [20] A. Haraux and M. A. Jendoubi, *The Convergence Problem for Dissipative Autonomous Systems*, Springer, 2015.
- [21] J. Morales and D. Poyato, *On the trend to global equilibrium for Kuramoto oscillators*, arXiv:1908.07657 (2019).
- [22] J. A. Carrillo, Y.-P. Choi, S.-Y. Ha, M.-J. Kang, and Y. Kim, *Contractivity of transport distances for the kinetic Kuramoto equation*, J. Stat. Phys. **156** (2014) 395–415.
- [23] M. Iacobelli, *A new perspective on Wasserstein distances for kinetic problems*, Arch. Ration. Mech. Anal. **244** (2022) 27–50.
- [24] C. Mouhot and C. Villani, *On Landau damping*, Acta Math. **207** (2011) 29–201.
- [25] J. Mierczyński, *Strictly cooperative systems with a first integral*, SIAM J. Math. Anal. **18**(3) (1987) 642–646.
- [26] C. Kuehn and P. Landi, *The mean-field Ott–Antonsen manifold is an unstable manifold in the continuum limit*, arXiv:2511.03833 (2025).
- [27] B. Pietras and A. Daffertshofer, *Ott–Antonsen attractiveness for parameter-dependent oscillatory networks*, Chaos **26** (2016) 103101.
- [28] E. Hille, *Ordinary Differential Equations in the Complex Domain*, Dover, 1997.
- [29] P. W. Bates and C. K. R. T. Jones, *Invariant manifolds for semilinear partial differential equations*, Dynamics Reported **2** (1989) 1–38.
- [30] S. H. Strogatz, *From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators*, Physica D **143** (2000) 1–20.
- [31] M. W. Hirsch, S. Smale and R. L. Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, 3rd ed., Academic Press, 2013. Theorem 8.2.1.